

EXACT CONVERGENCE CERTIFICATES AND ORDERING THEORY FOR OVERLAPPING ITERATIVE BLOCK MATRIX INVERSION

BINH T. NGUYEN*

Abstract. Iterative block matrix inversion (IBMI) approximates the full inverse of a symmetric positive definite matrix by repeated exact block-inverse updates. We develop an exact predictive and design theory for ordered overlapping sweeps. A complete sweep reduces to a return operator whose squared spectral radius is the exact worst-case asymptotic spectral factor of the full inverse-error operator. Our main result removes the apparent dependence of this return operator on the unknown inverse: every inter-complement transfer is computable directly from the original matrix through local principal-block solves. This gives an exact pre-iteration certificate $\rho(M_A)^2$ without forming A^{-1} . The certificate yields a two-block nested-overlap monotonicity theorem and an exact SPD example showing that the law already fails for three blocks. For singleton complements, the return factor becomes a weighted precision-cycle objective; exact rate-optimal ordering is NP-hard even for unit-diagonal SPD precision matrices. For a fixed block collection, local graph maps can be factored once and reused to score arbitrary sweep permutations exactly. Experimental results on covariance models and eight public SPD stiffness matrices validate the certificate against actual IBMI trajectories; a pre-frozen stress campaign shows ordering sensitivity ranging from negligible to substantial.

Key words. iterative block matrix inversion, symmetric positive definite matrix, overlapping block methods, return operator, subspace correction, matrix inverse approximation

MSC codes. 15A09, 15A18, 65F05, 65F10

1. Introduction. Iterative block matrix inversion (IBMI) [16] approximates the full inverse of an SPD matrix by exact block-inverse updates. Its original theory covers two nonoverlapping blocks, while multi-block experiments show sensitivity to overlap and ordering. We ask whether the asymptotic inverse-error factor can be determined from the matrix and ordered cover before iteration.

Let $A = A^\top \succ 0$, $H = A^{-1}$, and $E = X - H$. For an update block I with complement J , one IBMI update satisfies

$$E^+ = Q_I E Q_I^\top, \quad Q_I = H E_J H_{J,J}^{-1} E_J^\top.$$

The factor Q_I is an A -orthogonal projector. Products of such projections and their relation to subspace correction are classical [1, 19, 20].

For an ordered cover $I_1, \dots, I_K$, with $J_k = I_k^c$, the IBMI-specific structure appears after reducing a complete sweep to a complement-space return operator M . The nonzero sweep spectrum is exactly the nonzero spectrum of M , and the full inverse-error operator $\mathcal{L}(E) = Q E Q^\top$ satisfies

$$\rho(\mathcal{L}) = \rho(M)^2.$$

Written in inverse blocks, however, this representation appears post-hoc because it contains $H = A^{-1}$.

The central result removes that obstacle. For every update block I , complement J , and target index set L ,

$$(1.1) \quad H_{LJ} H_{J,J}^{-1} = E_L^\top E_I (-A_{I,I}^{-1} A_{I,J}) + E_L^\top E_J.$$

*Faculty of Mathematics and Computer Sciences, University of Science, Vietnam National University Ho Chi Minh City, Vietnam (ngtbinh@hcmus.edu.vn).

Therefore, every transition in the return cycle is available from a local principal-block solve with the original matrix A . Chaining these transfers gives an exact A -only return operator $M_A = M$ and hence $\rho(\mathcal{L}) = \rho(M_A)^2$. The contribution is the complete cyclic certificate and its consequences, not the isolated block identity in (1.1).

The certificate yields a two-block canonical-correlation law: nested update blocks cannot worsen the factor, while an exact SPD counterexample shows failure already for three blocks. For singleton complements, it becomes a weighted cycle of normalized precision couplings, giving finite-termination families and an NP-hard exact ordering problem. For fixed blocks, local graph maps can be cached and reused to score permutations exactly; we call budgeted search on these scores certificate-guided sweep ordering (CGSO), without claiming cheap arbitrary-cover search or universal acceleration.

Contributions and novelty boundary. The principal contribution of this work is an exact, A -only, pre-iteration return certificate for the asymptotic convergence factor of the full inverse-error operator. Building on this certificate, we establish (i) a nested-overlap monotonicity result for two blocks, together with an exact SPD counterexample showing that the analogous monotonicity can fail for three blocks; (ii) an exact precision-cycle formula for singleton complements and NP-hardness of the corresponding rate-optimal ordering problem; and (iii) a factor-once, score-many procedure for evaluating permutations of a fixed block collection. These results are supported numerically on structured covariance models, public SPD stiffness matrices, and controlled ordering-stress experiments.

Products of energy projections, fixed-cover convergence, and successive subspace-correction theory are classical [1, 19, 20]. Ordering effects in sequential correction methods have likewise been studied using greedy, randomized, and expected-rate analyses [10, 11, 14, 15]. Our contribution in this setting is the identification of the exact IBMI precision-cycle objective and the complexity of its rate-optimal ordering problem, together with exact reuse of the general-block certificate across permutations of a fixed block collection. The two-block analysis also draws on classical principal-angle and canonical-correlation geometry [2], while the Hamiltonian-cycle complexity result uses the classical hardness of the Hamiltonian cycle problem [8]. These ingredients yield, in the IBMI setting, explicit consequences for the return operator and its convergence factor.

Related work on optimized Schwarz methods, selected inversion, randomized inversion, and structured covariance approximation [6, 7, 13, 9, 18, 17, 5, 12] addresses related matrix-computational questions through different state variables or approximation mechanisms. The distinguishing feature of the present analysis is the exact A -computable IBMI return operator, which yields the square-law certificate for the full inverse-error iteration and enables the overlap and ordering results developed below.

2. IBMI and inverse-error geometry. Throughout, $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite and

$$H := A^{-1}.$$

For an index set $S \subseteq [n] := \{1, \dots, n\}$, E_S denotes the coordinate embedding whose columns are the standard basis vectors indexed by S . Thus, $H_{ST} = E_S^\top H E_T$. We use $\|x\|_A^2 = x^\top A x$ and write

$$V_I := \text{range}(E_I)$$

for the coordinate subspace supported on I .

91 **2.1. One exact block update.** Let $I \subsetneq [n]$ be nonempty and set $J = I^c$. After
 92 a simultaneous permutation of rows and columns, write

$$93 \quad (2.1) \quad A = \begin{bmatrix} A_{II} & A_{IJ} \\ A_{JI} & A_{JJ} \end{bmatrix}.$$

94 The approximate block-inversion step used by IBMI keeps the current approximation
 95 X_{JJ} on the complementary principal block and reconstructs the remaining blocks
 96 through the exact block-inverse relations [16]. With

$$97 \quad R_I := A_{II}^{-1} A_{IJ},$$

98 the update is

$$99 \quad (2.2) \quad \Phi_I(X_{JJ}) = \begin{bmatrix} A_{II}^{-1} + R_I X_{JJ} R_I^\top & -R_I X_{JJ} \\ -X_{JJ} R_I^\top & X_{JJ} \end{bmatrix}.$$

100 The same formula with $X_{JJ} = H_{JJ}$ is the exact inverse H .

101 **THEOREM 2.1** (IBMI inverse-error congruence). *Let X be any symmetric matrix*
 102 *and let $E = X - H$. One exact IBMI block update on I satisfies*

$$103 \quad (2.3) \quad E^+ = Q_I E Q_I^\top, \quad Q_I := H E_J H_{JJ}^{-1} E_J^\top.$$

104 *Equivalently, if*

$$105 \quad Z_I = \begin{bmatrix} -R_I \\ I \end{bmatrix}$$

106 *in the (I, J) ordering, then*

$$107 \quad E^+ = Z_I E_{JJ} Z_I^\top, \quad Q_I = Z_I E_J^\top.$$

108 *Proof.* Subtracting the exact block-inverse formula from (2.2) gives

$$109 \quad E^+ = \begin{bmatrix} R_I E_{JJ} R_I^\top & -R_I E_{JJ} \\ -E_{JJ} R_I^\top & E_{JJ} \end{bmatrix} = Z_I E_{JJ} Z_I^\top.$$

110 Since $E_{JJ} = E_J^\top E E_J$, it remains only to identify $Z_I E_J^\top$. The block equation $AH = I$
 111 gives

$$112 \quad A_{II} H_{IJ} + A_{IJ} H_{JJ} = 0,$$

113 hence

$$114 \quad H_{IJ} H_{JJ}^{-1} = -A_{II}^{-1} A_{IJ} = -R_I.$$

115 Therefore,

$$116 \quad H E_J H_{JJ}^{-1} = \begin{bmatrix} H_{IJ} \\ H_{JJ} \end{bmatrix} H_{JJ}^{-1} = \begin{bmatrix} -R_I \\ I \end{bmatrix} = Z_I,$$

117 which proves (2.3). □

118 Theorem 2.1 provides the structural starting point for the analysis: each IBMI up-
 119 date propagates the full inverse error through a two-sided congruence transformation,
 120 which forms the basis for the projection and sweep-level representations developed
 121 below.

2.2. Energy-projection structure.

PROPOSITION 2.2 (Energy projection). *For every nonempty proper block I ,*

$$(2.4) \quad Q_I^2 = Q_I, \quad Q_I^\top A = A Q_I.$$

Consequently, $P_I := I - Q_I$ is the A -orthogonal projector onto V_I , and Q_I is the A -orthogonal projector onto $V_I^\perp$.

Proof. Using $E_J^\top H E_J = H_{JJ}$,

$$Q_I^2 = H E_J H_{JJ}^{-1} (E_J^\top H E_J) H_{JJ}^{-1} E_J^\top = Q_I.$$

Moreover,

$$A Q_I = E_J H_{JJ}^{-1} E_J^\top,$$

which is symmetric, and therefore $Q_I^\top A = A Q_I$. Thus, Q_I is an A -orthogonal projector. Its nullspace contains V_I because $E_J^\top E_I = 0$, and both spaces have dimension $|I|$; hence $\ker(Q_I) = V_I$. Therefore, $P_I = I - Q_I$ is the A -orthogonal projector onto V_I and Q_I projects onto $V_I^\perp$. $\square$

Products of orthogonal projections and their interpretation as successive subspace corrections are classical; see Xu [19] and Xu–Zikatanov [20]. In the present setting, Proposition 2.2 provides the structural link between the IBMI error update and this classical framework through the A -orthogonal projection representation.

2.3. Ordered sweeps and fixed-cover convergence. Let $\mathcal{I} = (I_1, \dots, I_K)$ be an ordered family of nonempty proper blocks and define $Q := Q_{I_K} \cdots Q_{I_1}$. One complete sweep therefore maps the inverse error as

$$(2.5) \quad E^{(r+1)} = Q E^{(r)} Q^\top,$$

and repeated fixed-order sweeps satisfy

$$(2.6) \quad E^{(r)} = Q^r E^{(0)} (Q^\top)^r.$$

COROLLARY 2.3 (General fixed-cover convergence). *Assume that*

$$(2.7) \quad \bigcup_{k=1}^K I_k = [n].$$

Then,

$$\|Q\|_A < 1, \quad \rho(Q) < 1,$$

and, for every symmetric initial approximation $X^{(0)}$,

$$X^{(r)} \longrightarrow H = A^{-1}.$$

No disjointness assumption is required.

Proof. Each Q_{I_k} is an A -orthogonal projector and is therefore nonexpansive in the A -norm. Suppose that $\|Qx\|_A = \|x\|_A$ for some $x \neq 0$. The chain

$$\|Q_{I_K} \cdots Q_{I_1} x\|_A \leq \cdots \leq \|Q_{I_1} x\|_A \leq \|x\|_A$$

must then contain equality at every step. Equality for an orthogonal projection holds exactly on its range. Working successively through the factors shows that x belongs to

$$\bigcap_{k=1}^K V_{I_k}^{\perp A} = \left(\sum_{k=1}^K V_{I_k} \right)^{\perp A}.$$

The cover condition (2.7) implies $\sum_k V_{I_k} = \mathbb{R}^n$, so the intersection is $\{0\}$, a contradiction. Compactness of the A -unit sphere gives $\|Q\|_A < 1$, hence $\rho(Q) < 1$. Equation (2.6) then gives $E^{(r)} \rightarrow 0$. $\square$

Corollary 2.3 establishes convergence for any fixed multi-block cover by identifying the IBMI error iteration with a product of energy projections. The convergence argument itself follows classical subspace-correction theory, while the IBMI-specific ingredient is the congruence representation in (2.3) that enables this identification. This result guarantees convergence but does not characterize its rate, which can depend strongly on the overlap pattern and index geometry. The next section exploits the block structure in (2.3) to derive a reduced representation of the complete sweep dynamics and an exact characterization of the associated asymptotic convergence factor.

3. Return-operator spectral theory. The projection representation in Section 2 proves convergence, but by itself it does not expose the quantity that controls the rate of an ordered overlapping sweep. We now exploit the special coordinate structure of the IBMI projectors to compress the nonzero sweep dynamics to a return operator acting on the complement of the first block.

Let $\mathcal{I} = (I_1, \dots, I_K)$ be a fixed ordered family of nonempty proper blocks, set $J_k = I_k^c$, and write

$$Q_k := Q_{I_k} = H E_{J_k} H_{J_k J_k}^{-1} E_{J_k}^\top, \quad Q := Q_K \cdots Q_1.$$

Define the transfer product

$$(3.1) \quad B := H_{J_K J_K}^{-1} H_{J_K J_{K-1}} H_{J_{K-1} J_{K-1}}^{-1} \cdots H_{J_2 J_2}^{-1} H_{J_2 J_1} H_{J_1 J_1}^{-1}.$$

Thus, $B \in \mathbb{R}^{|J_K| \times |J_1|}$. The reduced return operator is

$$(3.2) \quad M := H_{J_1 J_K} B \in \mathbb{R}^{|J_1| \times |J_1|}.$$

The adjective “return” refers to the fact that the sweep first transfers an error state from the initial complement J_1 through the ordered sequence of complements and then closes the transfer from J_K back to J_1 through $H_{J_1 J_K}$.

THEOREM 3.1 (Return-operator factorization). *For every ordered IBMI sweep,*

$$(3.3) \quad Q = H E_{J_K} B E_{J_1}^\top.$$

If $U := H E_{J_K} B$ and $V := E_{J_1}$, then $Q = UV^\top$, $M = V^\top U$, and for every integer $s \geq 1$,

$$(3.4) \quad Q^s = U M^{s-1} V^\top = H E_{J_K} B M^{s-1} E_{J_1}^\top.$$

Consequently, the classical $UV^\top - V^\top U$ relation gives

$$(3.5) \quad \det(\lambda I_n - Q) = \lambda^{n-|J_1|} \det(\lambda I_{|J_1|} - M),$$

and hence,

$$(3.6) \quad \sigma(Q) \setminus \{0\} = \sigma(M) \setminus \{0\}$$

with algebraic multiplicities. In particular,

$$(3.7) \quad \rho(Q) = \rho(M).$$

Proof. Multiplying two consecutive projectors gives

$$Q_{k+1}Q_k = HE_{J_{k+1}}H_{J_{k+1}J_{k+1}}^{-1}H_{J_{k+1}J_k}H_{J_kJ_k}^{-1}E_{J_k}^\top.$$

Iterating this identity from $k = 1$ to $K - 1$ yields (3.3) with B as in (3.1). The definitions of U and V then give $Q = UV^\top$ and

$$V^\top U = E_{J_1}^\top HE_{J_K}B = H_{J_1J_K}B = M.$$

Sylvester's determinant identity, applied to $\lambda I_n - UV^\top$, gives (3.5) as a polynomial identity. This is the classical $UV^\top - V^\top U$ spectral relation; here it is applied to the IBMI-specific factorization above. Equations (3.6) and (3.7) follow immediately. Finally, $Q^1 = UV^\top$, and if $Q^s = UM^{s-1}V^\top$, then

$$Q^{s+1} = UM^{s-1}(V^\top U)V^\top = UM^sV^\top,$$

which proves (3.4) by induction. $\square$

Theorem 3.1 provides a structural reduction specific to the ordered Schur complement reconstruction underlying IBMI. While the spectral correspondence between $UV^\top$ and $V^\top U$ is classical, the theorem identifies the explicit return matrix M in (3.2) and shows that it captures all nonzero eigenvalues of the full sweep operator Q . Thus, the persistent spectral dynamics of a complete IBMI sweep are determined by M . When the first complement J_1 is small, this representation yields a substantial dimensional reduction; no such reduction is implied when $|J_1|$ is comparable to n .

3.1. Exact inverse-error dynamics and the square law. Let $\mathcal{L} : \mathbb{S}^n \rightarrow \mathbb{S}^n$ denote one complete sweep of the inverse error,

$$(3.8) \quad \mathcal{L}(E) := QEQ^\top.$$

Combining Theorem 3.1 with (2.6) gives an exact reduced expression for every sweep.

COROLLARY 3.2 (Exact return dynamics). *For every $s \geq 1$ and every symmetric initial error $E^{(0)}$,*

$$(3.9) \quad E^{(s)} = UM^{s-1}V^\top E^{(0)}V(M^\top)^{s-1}U^\top.$$

Thus, after the first sweep, all persistent inverse-error dynamics are mediated by powers of M and $M^\top$.

Proof. Insert (3.4) and its transpose into $E^{(s)} = Q^s E^{(0)} (Q^\top)^s$. $\square$

THEOREM 3.3 (Full inverse-error square law). *For the real linear operator $\mathcal{L}$ in (3.8), restricted to symmetric matrices,*

$$(3.10) \quad \rho(\mathcal{L}) = \rho(Q)^2 = \rho(M)^2.$$

Hence, the worst-case asymptotic spectral factor of a fixed IBMI sweep on the full inverse error is the square of the return spectral radius.

Proof. On the full matrix space, vectorization gives

$$\text{vec}(\mathcal{L}(E)) = (Q \otimes Q) \text{vec}(E),$$

so the spectral radius of the unrestricted operator is $\rho(Q)^2$. Because $\mathbb{S}^n$ is invariant under $E \mapsto QEQ^\top$, the spectral radius of the restriction is at most $\rho(Q)^2$.

For the reverse inequality, complexify the real symmetric matrix space. Let λ be an eigenvalue of Q with $|\lambda| = \rho(Q)$ and let $v \neq 0$ satisfy $Qv = \lambda v$ over $\mathbb{C}$. The nonzero matrix $vv^\top$ is complex symmetric and

$$\mathcal{L}_{\mathbb{C}}(vv^\top) = (Qv)(Qv)^\top = \lambda^2 vv^\top.$$

Thus, the complexified restriction has spectral radius at least $|\lambda|^2 = \rho(Q)^2$. A real finite-dimensional operator and its complexification have the same spectral radius, proving $\rho(\mathcal{L}) = \rho(Q)^2$. Equation (3.7) completes the proof. $\square$

Theorem 3.3 characterizes the asymptotic convergence factor of the error operator, not necessarily that of every individual trajectory. In particular, an initial error $E^{(0)}$ may have no component in a dominant invariant mode and may therefore converge at a strictly faster asymptotic rate. This distinction is relevant both to the interpretation of the numerical experiments and to the finite-termination results below.

3.2. Finite-time bounds and nonnormal return dynamics. The return spectrum determines the asymptotic factor. For nonnormal M , however, $\rho(M)$ alone need not describe preasymptotic behavior. The exact dynamics give the following bound without any diagonalizability assumption.

PROPOSITION 3.4 (Finite-time return bound). *For every $s \geq 1$,*

$$(3.11) \quad \|E^{(s)}\|_2 \leq \|U\|_2^2 \|M^{s-1}\|_2^2 \|E^{(0)}\|_2.$$

If $M = X\Lambda X^{-1}$ is diagonalizable, then

$$(3.12) \quad \|E^{(s)}\|_2 \leq \|U\|_2^2 \kappa_2(X)^2 \rho(M)^{2(s-1)} \|E^{(0)}\|_2.$$

More generally, if the largest Jordan block associated with eigenvalues of modulus $\rho(M) > 0$ has size p , then

$$(3.13) \quad \|E^{(s)}\|_2 = O\left(s^{2(p-1)} \rho(M)^{2s}\right)$$

up to constants independent of s .

Proof. Since $V = E_{J_1}$ has orthonormal columns, $\|V\|_2 = 1$. Taking norms in (3.9) proves (3.11). The standard bound $\|M^k\|_2 \leq \kappa_2(X)\rho(M)^k$ gives (3.12). The Jordan canonical form gives $\|M^k\| = O(k^{p-1}\rho(M)^k)$; squaring this factor in (3.11) yields (3.13), with an inessential shift from s to $s-1$ absorbed into the constant. $\square$

This separates two roles of the return operator: $\rho(M)^2$ is the exact asymptotic operator factor, while $\|M^k\|$ controls finite-sweep transients. The distinction motivates the nonnormal stress tests in Section 9.

4. Finite and near-finite convergence. The return representation also identifies when geometric convergence collapses to exact finite termination. We first state the operator-level result and then quantify perturbations away from nilpotency.

4.1. Nilpotent return dynamics. For a square matrix T , write

$$\nu(T) := \min\{m \geq 1 : T^m = 0\}$$

when T is nilpotent. For a particular initial inverse error, define the trajectory termination index

$$\nu(Q, E^{(0)}) := \min\{m \geq 1 : Q^m E^{(0)} (Q^\top)^m = 0\},$$

whenever such an m exists.

THEOREM 4.1 (Finite termination through the return operator). *For every $p \geq 1$,*

$$(4.1) \quad Q^{p+1} = 0 \iff BM^p = 0.$$

Consequently, if $M^p = 0$, then $Q^{p+1} = 0$ and IBMI is exact after at most $p + 1$ complete sweeps for every symmetric initial approximation:

$$(4.2) \quad E^{(p+1)} = 0.$$

If, in addition, B has full column rank, then

$$(4.3) \quad Q^{p+1} = 0 \iff M^p = 0.$$

Finally, the inverse-error operator satisfies

$$(4.4) \quad \mathcal{L}^s = 0 \text{ on } \mathbb{S}^n \iff Q^s = 0.$$

Proof. Equation (3.4) gives $Q^{p+1} = HE_{J_K} BM^p E_{J_1}^\top$. The matrix HE_{J_K} has full column rank and $E_{J_1}^\top$ has full row rank; multiplying by a left inverse of the former and a right inverse of the latter proves (4.1). The implication $M^p = 0 \Rightarrow E^{(p+1)} = 0$ follows immediately. If B has full column rank, then $BM^p = 0$ is equivalent to $M^p = 0$, proving (4.3).

If $Q^s = 0$, then clearly $\mathcal{L}^s = 0$. Conversely, if $\mathcal{L}^s = 0$ on all symmetric matrices, apply it to I to obtain $Q^s (Q^s)^\top = 0$, which implies $Q^s = 0$. $\square$

The exact condition without a rank assumption is therefore $BM^p = 0$; nilpotency of M is sufficient but need not be necessary when B loses rank. Likewise, operator finite termination should not be confused with termination of one special trajectory. Whenever Q is nilpotent,

$$\nu(Q, E^{(0)}) \leq \nu(Q),$$

and strict inequality can occur when the initial error fails to excite a surviving mode.

The simplest finite regime is $M = 0$. Then $Q^2 = 0$ and every inverse error vanishes after two sweeps. The factorization $M = H_{J_1 J_K} B$ gives a direct precision-decoupling condition.

COROLLARY 4.2 (Endpoint precision decoupling). *If*

$$(4.5) \quad H_{J_1 J_K} = 0,$$

then $M = 0$, $Q^2 = 0$, and $E^{(2)} = 0$ for every symmetric initial error. If $B \neq 0$, then the termination is genuinely two-sweep at the operator level: $Q \neq 0$ and $Q^2 = 0$.

Proof. Equation (3.2) gives $M = 0$, and Theorem 4.1 gives $Q^2 = 0$. If $B \neq 0$, then $HE_{J_K}B \neq 0$ because HE_{J_K} has full column rank; multiplication by the full-row-rank matrix $E_{J_1}^\top$ preserves nonzeroness here, so $Q = HE_{J_K}BE_{J_1}^\top \neq 0$. $\square$

Condition (4.5) is a zero pattern in the precision matrix $H = A^{-1}$; precision sparsity itself is classical. The consequence here is specific to the ordered IBMI sweep: the endpoint zero closes the return loop and forces exact two-sweep termination. Section 8 makes this mechanism explicit for singleton complements.

4.2. Near-nilpotency and quadratic inverse-error collapse. Exact zeros are fragile, so a useful theory should also describe perturbations of nilpotent return dynamics. The next result gives a nonasymptotic estimate.

THEOREM 4.3 (Near-nilpotent return bound). *Suppose*

$$(4.6) \quad M = N + \Delta, \quad N^p = 0, \quad \|\Delta\|_2 \leq \varepsilon,$$

and assume

$$(4.7) \quad \max\{\|M\|_2, \|N\|_2\} \leq L.$$

Then,

$$(4.8) \quad \|M^p\|_2 \leq pL^{p-1}\varepsilon$$

and, for every symmetric initial error,

$$(4.9) \quad \|E^{(p+1)}\|_2 \leq \|U\|_2^2 p^2 L^{2p-2} \varepsilon^2 \|E^{(0)}\|_2.$$

Thus, a first-order perturbation of a nilpotent return produces a second-order bound on the full inverse error after the corresponding collapse horizon.

Proof. The telescoping identity

$$M^p - N^p = \sum_{j=0}^{p-1} M^{p-1-j}(M - N)N^j$$

together with $N^p = 0$, (4.6), and (4.7) yields

$$\|M^p\|_2 \leq \sum_{j=0}^{p-1} L^{p-1-j} \varepsilon L^j = pL^{p-1} \varepsilon.$$

Corollary 3.2 at $s = p + 1$ gives

$$E^{(p+1)} = UM^pV^\top E^{(0)}V(M^\top)^pU^\top.$$

Since $\|V\|_2 = 1$, taking spectral norms and inserting (4.8) proves (4.9). $\square$

For a parameterized family, the $O(\varepsilon^2)$ interpretation of (4.9) requires $\|U\|_2$ and L to remain uniformly bounded as $\varepsilon \rightarrow 0$. This condition prevents the prefactor from hiding a singular dependence on the perturbation parameter.

COROLLARY 4.4 (Approximate endpoint decoupling). *Let*

$$C := H_{J_1 J_K}, \quad M = CB.$$

If $\|C\|_2 \leq \varepsilon$, then

$$(4.10) \quad \rho(M) \leq \|M\|_2 \leq \varepsilon \|B\|_2$$

and therefore,

$$(4.11) \quad \rho(\mathcal{L}) = \rho(M)^2 \leq \varepsilon^2 \|B\|_2^2.$$

Moreover,

$$(4.12) \quad \|E^{(2)}\|_2 \leq \|U\|_2^2 \|B\|_2^2 \varepsilon^2 \|E^{(0)}\|_2.$$

Proof. The first bound follows from $M = CB$ and submultiplicativity. The square-law bound (4.11) follows from Theorem 3.3. Finally, Corollary 3.2 with $s = 2$ gives $E^{(2)} = U M V^\top E^{(0)} V M^\top U^\top$; use $\|M\|_2 \leq \varepsilon \|B\|_2$. $\square$

Corollary 4.4 identifies a precise mechanism for near-finite convergence. When the endpoint precision coupling is of order ε , the spectral radius of the return operator can be of first order in ε , whereas the corresponding asymptotic spectral factor for the full inverse-error iteration is of second order. However, overlap modifies not only the closing block C but also the transfer factors comprising B . Thus, increasing overlap need not decrease the convergence factor for a general ordered multi-block cover. For a fixed ordered cover, the exact asymptotic spectral factor is

$$(4.13) \quad \rho(H_{J_1 J_K} H_{J_K J_K}^{-1} H_{J_K J_{K-1}} \cdots H_{J_2 J_2}^{-1} H_{J_2 J_1} H_{J_1 J_1}^{-1})^2.$$

This dependence motivates the matched-overlap experiments in Section 9, in which the block cardinalities and overlap incidence are held fixed while the return geometry is varied.

5. An exact A -only return certificate. The return operator derived in Section 3 is expressed in terms of blocks of $H = A^{-1}$. Although this representation provides an exact characterization of the sweep dynamics, its direct evaluation would require access to the inverse matrix that IBMI seeks to approximate. The following result eliminates this dependence on H and yields an equivalent representation computable directly from A and the prescribed block structure.

For a nonempty proper block I , write $J = I^c$. For an arbitrary index set $L \subseteq [n]$, define the transfer from the complement J to L by

$$(5.1) \quad \mathcal{R}_{L \leftarrow J} := H_{LJ} H_{JJ}^{-1}.$$

THEOREM 5.1 (A -only local-transfer identity). *For every nonempty proper I , with $J = I^c$, and every $L \subseteq [n]$,*

$$(5.2) \quad \mathcal{R}_{L \leftarrow J} = E_L^\top E_I (-A_{II}^{-1} A_{IJ}) + E_L^\top E_J.$$

Thus, the transfer (5.1) is computable from A and the index sets by solving with the principal block A_{II} ; the full inverse H need not be formed.

Proof. The block equation $AH = I$ gives $A_{II} H_{IJ} + A_{IJ} H_{JJ} = 0$, and hence,

$$H_{IJ} H_{JJ}^{-1} = -A_{II}^{-1} A_{IJ}.$$

Together with $H_{JJ} H_{JJ}^{-1} = I_J$, this yields

$$H E_J H_{JJ}^{-1} = E_I (-A_{II}^{-1} A_{IJ}) + E_J.$$

Left multiplication by $E_L^\top$ proves (5.2). $\square$

Theorem 5.1 provides the local identities needed to construct the return dynamics directly from A . Although each individual identity follows from standard block equations, composing these local maps over an ordered cover yields an exact A -only representation of the persistent complete-sweep dynamics. This composition leads directly to the pre-iteration certificate developed next.

THEOREM 5.2 (Exact pre-iteration return certificate). *Let $\mathcal{I} = (I_1, \dots, I_K)$ be an ordered IBMI cover and $J_k = I_k^c$. Define*

$$\mathcal{R}_{k+1,k} := \mathcal{R}_{J_{k+1} \leftarrow J_k}, \quad k = 1, \dots, K-1, \quad \mathcal{R}_{1,K} := \mathcal{R}_{J_1 \leftarrow J_K},$$

where every transfer is evaluated by (5.2). Then,

$$(5.3) \quad M_A := \mathcal{R}_{1,K} \mathcal{R}_{K,K-1} \cdots \mathcal{R}_{2,1} = M.$$

Consequently, the full inverse-error sweep operator satisfies

$$(5.4) \quad \rho(\mathcal{L}) = \rho(M_A)^2.$$

Hence, $\rho(M_A)^2$ is an exact asymptotic operator certificate available from A and the ordered cover before the inverse is formed.

Proof. By definition and Theorem 5.1, $\mathcal{R}_{J_{k+1} \leftarrow J_k} = H_{J_{k+1} J_k} H_{J_k}^{-1}$. Multiplying the transfers in cyclic order reproduces exactly the return product in (3.2). Equation (5.4) then follows from Theorem 3.3. $\square$

REMARK 5.1 (Asymptotic versus finite-sweep behavior). Equation (5.4) gives an exact asymptotic spectral characterization of the error operator. It does not, however, eliminate the nonnormal transient effects described in Proposition 3.4. Finite-sweep convergence depends on the behavior of the powers M_A^s , and hence, on $\|M_A^s\|$, rather than on the spectral-radius prediction $\rho(M_A)^s$ alone.

5.1. Cost of the certificate. Let $b_k = |I_k|$ and $r_k = |J_k|$. With a dense Cholesky factorization, forming the local graph map

$$T_k := H E_{J_k} H_{J_k}^{-1} = E_{I_k} (-A_{I_k I_k}^{-1} A_{I_k J_k}) + E_{J_k}$$

requires, to leading order, approximately

$$(5.5) \quad \frac{1}{3} b_k^3 + 2 b_k^2 r_k$$

floating-point operations for the factorization and triangular solves. The cyclic return product can then be assembled using only restrictions of these local maps and reduced matrix products. Consequently, the certificate can be evaluated without forming the full n -by- n inverse. This structural reduction does not by itself imply a lower computational cost than one IBMI sweep when all required factorizations must be computed from scratch. It is particularly advantageous when the complements are small or when the local factorizations are already available and can be reused.

6. Overlap laws and their boundary. The A -only certificate turns overlap from a scalar design parameter into a spectral question. The principal-angle machinery used below is classical [2]; the IBMI-specific statement is the exact identification of that geometry with the full inverse-error factor. For two blocks this question has a monotonic answer.

THEOREM 6.1 (Two-block nested-overlap monotonicity). *For a two-block sweep with complements J_1, J_2 , let*

$$(6.1) \quad C := H_{J_1 J_1}^{-1/2} H_{J_1 J_2} H_{J_2 J_2}^{-1/2}.$$

Then,

$$(6.2) \quad \rho(M) = \|C\|_2^2, \quad \rho(\mathcal{L}) = \|C\|_2^4.$$

If $I_k \subseteq I'_k$ for $k = 1, 2$, equivalently $J'_k \subseteq J_k$, then

$$(6.3) \quad \rho(M') \leq \rho(M), \quad \rho(\mathcal{L}') \leq \rho(\mathcal{L}).$$

Proof. For $K = 2$, $M = H_{J_1 J_2} H_{J_2 J_2}^{-1} H_{J_2 J_1} H_{J_1 J_1}^{-1}$, which is similar to $CC^\top$ and proves (6.2). For the monotonicity statement, define

$$U_k := H^{1/2} E_{J_k} H_{J_k J_k}^{-1/2}.$$

The columns of U_k are orthonormal and $U_1^\top U_2 = C$. Hence, $\|C\|_2$ is the largest canonical correlation, equivalently the maximum absolute inner product between unit vectors in

$$\mathcal{S}_k := \text{range}(H^{1/2} E_{J_k}).$$

If $J'_k \subseteq J_k$, then $\mathcal{S}'_k \subseteq \mathcal{S}_k$. The variational domain defining the largest canonical correlation therefore only shrinks:

$$\max_{\substack{x \in \mathcal{S}'_1, y \in \mathcal{S}'_2 \\ \|x\|_2 = \|y\|_2 = 1}} |x^\top y| \leq \max_{\substack{x \in \mathcal{S}_1, y \in \mathcal{S}_2 \\ \|x\|_2 = \|y\|_2 = 1}} |x^\top y|.$$

Thus, $\|C'\|_2 \leq \|C\|_2$, and (6.3) follows. $\square$

The two-block monotonicity result does not extend unconditionally beyond two blocks. The following exact SPD construction shows that monotonicity can already fail for a three-block ordered cover.

THEOREM 6.2 (Three-block nonmonotonicity). *There exists an SPD matrix and an ordered three-block cover for which enlarging one update block strictly increases the return spectral radius.*

Proof. Let

$$H = \frac{1}{100} \begin{bmatrix} 784 & -163 & -265 & -24 & -357 & -115 \\ -163 & 598 & 192 & 101 & -427 & 235 \\ -265 & 192 & 260 & 34 & 5 & 159 \\ -24 & 101 & 34 & 855 & 113 & 197 \\ -357 & -427 & 5 & 113 & 842 & -17 \\ -115 & 235 & 159 & 197 & -17 & 217 \end{bmatrix}.$$

Its leading principal minors are

$$\frac{196}{25}, \quad \frac{442263}{10000}, \quad \frac{30339667}{500000}, \quad \frac{50841748123}{10^8}, \quad \frac{4880230217611}{5 \cdot 10^9}, \quad \frac{232688511158651}{10^{12}},$$

all positive, so $H \succ 0$. Take

$$J_1 = \{1, 3\}, \quad J_2 = \{2, 3, 6\}, \quad J_3 = \{4, 5\},$$

and enlarge the middle update block by replacing J_2 with $J'_2 = \{2, 6\} \subset J_2$. Exact rational arithmetic gives

$$p_{\text{before}}(x) = \frac{167100723926068535x^2 - 1338568090325562x + 1688270156316}{167100723926068535},$$

whereas

$$p_{\text{after}}(x) = \frac{2347659967233605x^2 - 29322446429644x - 1266602954844}{2347659967233605}.$$

The strict comparison can be certified without floating-point eigenvalues. Let the numerators of the two displayed polynomials be q_{before} and q_{after} . Direct integer arithmetic gives

$$q_{\text{before}}(0) > 0, \quad q_{\text{before}}\left(\frac{1}{500}\right) < 0, \quad q_{\text{before}}\left(\frac{1}{100}\right) > 0.$$

Because q_{before} is quadratic with positive leading coefficient, these two sign changes isolate its two real roots in $(0, \frac{1}{500})$ and $(\frac{1}{500}, \frac{1}{100})$, respectively. Hence, $\rho(M_{\text{before}}) < \frac{1}{100}$. For the enlarged middle block,

$$q_{\text{after}}\left(-\frac{1}{50}\right) > 0, \quad q_{\text{after}}(0) < 0, \quad q_{\text{after}}\left(\frac{1}{50}\right) < 0.$$

Since q_{after} has positive leading coefficient and negative constant term, it has one negative and one positive root. The first two signs place the negative root in $(-1/50, 0)$, while $q_{\text{after}}(1/50) < 0$ forces the positive root to lie in $(1/50, \infty)$. Therefore,

$$\rho(M_{\text{after}}) > \frac{1}{50} > \frac{1}{100} > \rho(M_{\text{before}}).$$

Thus, enlarging the middle update block strictly worsens the exact return spectral radius. Numerically,

$$\rho(M_{\text{before}}) \approx 0.0064422599637, \quad \rho(M_{\text{after}}) \approx 0.0302974317480. \quad \square$$

REMARK 6.1. *Theorem 6.2 establishes that the two-block monotonicity law can fail already for $K = 3$. The construction is specific to the three-block setting and, without an additional lifting argument, does not by itself establish failure for every $K \geq 3$.*

7. Ordering structure and certificate-guided design. For singleton complements, the return certificate becomes an exact weighted cycle. Let $J_i = \{i\}$, so $I_i = [n] \setminus \{i\}$.

THEOREM 7.1 (Singleton precision-cycle formula). *For an ordered sequence $\pi = (i_1, \dots, i_s)$,*

$$Q_\pi = \frac{\prod_{r=1}^{s-1} h_{i_{r+1}i_r}}{\prod_{r=1}^s h_{i_r i_r}} H e_{i_s} e_{i_1}^\top.$$

For a permutation of $[n]$, the only possible nonzero eigenvalue is

$$(7.1) \quad \lambda_\pi = \frac{h_{i_1 i_n} \prod_{r=1}^{n-1} h_{i_{r+1} i_r}}{\prod_{r=1}^n h_{i_r i_r}}, \quad \rho(\mathcal{L}_\pi) = |\lambda_\pi|^2.$$

476 *Equivalently, with $c_{ij} = h_{ij}/\sqrt{h_{ii}h_{jj}}$,*

$$477 \quad |\lambda_\pi| = |c_{i_1 i_n}| \prod_{r=1}^{n-1} |c_{i_{r+1} i_r}|.$$

478 *Proof.* For a singleton complement, $Q_i = H e_i e_i^\top / h_{ii}$. Successive multiplication
 479 produces one factor $e_j^\top H e_i = h_{ji}$ at each transition. Closing the rank-one product
 480 produces the endpoint factor $h_{i_1 i_n}$. $\square$

481 **COROLLARY 7.2** (Path-cycle dichotomy). *If all consecutive couplings along π*
 482 *are nonzero, then $Q_\pi \neq 0$ and*

$$483 \quad Q_\pi^2 = 0 \iff h_{i_1 i_n} = 0.$$

484 *Thus, an ordered Hamiltonian path in the precision graph whose endpoints are not*
 485 *adjacent yields genuine two-sweep exactness.*

486 The singleton return space is one-dimensional. Hence, any finite-terminating
 487 singleton-complement sweep has nilpotency depth at most two; higher finite-sweep
 488 indices, if realizable, require a multidimensional return space.

489 Hamiltonian Circuit remains NP-complete even under strong graph restrictions
 490 [8]; we use it only as the source problem for the following IBMI-specific reduction.

491 **THEOREM 7.3** (NP-hard exact singleton ordering). *Consider the decision prob-*
 492 *lem whose input is a rational SPD matrix A and a rational threshold $\tau > 0$, and*
 493 *which asks whether some singleton-complement sweep order π satisfies $\rho(\mathcal{L}_\pi) \leq \tau$.*
 494 *This problem is NP-hard, even on instances for which the associated precision matrix*
 495 *$H = A^{-1}$ has unit diagonal.*

496 *Proof.* Reduce from undirected Hamiltonian Cycle. Given $G = (V, E)$ with $n =$
 497 $|V|$, set $\varepsilon = 1/(4n)$ and define the rational unit-diagonal matrix

$$498 \quad h_{ij} = \begin{cases} \varepsilon, & \{i, j\} \in E, \\ 2\varepsilon, & \{i, j\} \notin E, \end{cases} \quad i \neq j.$$

499 The off-diagonal absolute row sum is at most $2\varepsilon(n-1) = (n-1)/(2n) < 1$, so H
 500 is SPD by strict diagonal dominance. Its entries have $O(\log n)$ bits. Set $A = H^{-1}$.
 501 By the adjugate formula (equivalently, exact rational Gaussian elimination), every
 502 entry of A is a ratio of determinants of rational matrices of order at most n ; standard
 503 determinant bit bounds therefore give a representation of A with polynomial encoding
 504 length, computable in polynomial time.

505 Because $h_{ii} = 1$, the full inverse-error factor is

$$506 \quad \rho(\mathcal{L}_\pi) = |\lambda_\pi|^2 = \prod_{\{i,j\} \text{ on the cycle induced by } \pi} h_{ij}^2.$$

507 If G has a Hamiltonian cycle, its corresponding order has factor exactly ε^{2n} . Con-
 508 versely, if the induced cycle contains a nonedge, its factor is at least

$$509 \quad (2\varepsilon)^2 \varepsilon^{2(n-1)} = 4\varepsilon^{2n}.$$

510 Set the rational threshold $\tau = \varepsilon^{2n}$. Its encoding length is $O(n \log n)$. The constructed
 511 IBMI instance is therefore a YES instance if and only if G has a Hamiltonian cycle,
 512 giving a polynomial-time many-one reduction. $\square$

Theorem 7.3 concerns the ordering problem induced specifically by the IBMI singleton return cycle. Ordering effects, adaptive strategies, and randomized re-orderings are well established in the literature on multiplicative Schwarz, subspace-correction, SOR, and related methods [10, 11, 14, 15]. The contribution here is the exact precision-cycle objective arising from the IBMI return representation and the complexity characterization of its rate-optimal ordering problem.

7.1. Factor once, score many. For a fixed block collection, Theorem 5.1 yields a low-cost reuse mechanism.

PROPOSITION 7.4 (Cached permutation scoring). *Fix update blocks $\{I_1, \dots, I_K\}$ and form once*

$$T_k = E_{I_k}(-A_{I_k I_k}^{-1} A_{I_k J_k}) + E_{J_k}.$$

For any permutation π of the fixed blocks,

$$\mathcal{R}_{J_{\pi_{r+1}} \leftarrow J_{\pi_r}} = E_{J_{\pi_{r+1}}}^\top T_{\pi_r}.$$

Hence, after the K local graph maps are available, an arbitrary permutation can be scored exactly through reduced restrictions, products, and $\rho(M_A)$, with no additional solve involving $A_{I_k I_k}$.

Proof. Apply Theorem 5.1 with source complement J_{π_r} and target $L = J_{\pi_{r+1}}$. $\square$

We refer to budgeted search over these cached exact scores as *certificate-guided sweep ordering* (CGSO). The NP-hardness theorem precludes interpreting a generic local search as an exact global optimizer. For singleton complements, however, the log-cycle objective has a particularly cheap symmetric 2-opt update: replacing boundary edges $(a, b), (c, d)$ by $(a, c), (b, d)$ changes only

$$\Delta = w_{ac} + w_{bd} - w_{ab} - w_{cd}, \quad w_{ij} := -\log |c_{ij}|,$$

so each proposal is scored in $O(1)$ time once the weights are formed.

8. Precision-graph consequences for finite termination. The singleton precision-cycle formula and path-cycle dichotomy were proved in Section 7. Precision sparsity itself is classical [4]; the IBMI-specific statement is how an ordered precision path closes, or fails to close, the return loop.

8.1. An arbitrary-dimensional exact family. The path-cycle criterion is not a low-dimensional accident. It yields a family of genuine two-sweep examples in every dimension $n \geq 3$.

COROLLARY 8.1 (Arbitrary-dimensional genuine two-sweep family). *Let $n \geq 3$ and let $H \in \mathbb{R}^{n \times n}$ be any SPD tridiagonal matrix whose first sub- and super-diagonal entries are all nonzero:*

$$(8.1) \quad H = \begin{bmatrix} d_1 & a_1 & & & \\ a_1 & d_2 & a_2 & & \\ & a_2 & \ddots & \ddots & \\ & & \ddots & d_{n-1} & a_{n-1} \\ & & & a_{n-1} & d_n \end{bmatrix} \succ 0, \quad a_r \neq 0.$$

Set $A = H^{-1}$ and use the singleton-complement blocks

$$I_r = [n] \setminus \{r\}, \quad r = 1, \dots, n,$$

in the order $1, 2, \dots, n$. Then, the complete-sweep propagator satisfies

$$(8.2) \quad Q \neq 0, \quad Q^2 = 0.$$

Hence, IBMI is exact after two complete sweeps for every symmetric initial approximation, but not after one sweep for every initial approximation.

Proof. The tridiagonal zero pattern gives $h_{r+1,r} = a_r \neq 0$ for $r = 1, \dots, n-1$, while $h_{1n} = 0$ because $n \geq 3$. Apply Corollary 7.2 to the permutation $(1, 2, \dots, n)$. The final statement follows because $Q \neq 0$, so the operator $E \mapsto QEQ^\top$ is nonzero, whereas $Q^2 = 0$ makes its second iterate zero. $\square$

The finite-termination mechanism should be distinguished from general sparse-inverse structure. Precision sparsity or chordality alone does not imply finite termination. For example, the path graph in Corollary 8.1 is chordal, whereas a complete precision graph is also chordal and generically has no missing closing edge. The relevant structure here is an *ordered open precision path with endpoint decoupling*. More generally, (7.1) shows that the effect of ordering is determined by a weighted return-cycle product and therefore cannot be characterized by graph topology alone.

For singleton complements, the return space is one-dimensional, and the return matrix is therefore scalar. A nilpotent scalar return must vanish, so Theorem 4.1 implies that finite termination, when it occurs, requires at most two sweeps. Consequently, any finite-termination mechanism with a higher nilpotency index must involve a multidimensional return space. Whether such higher-index behavior is realizable within the common-SPD coordinate structure is not required for the results developed here.

9. Numerical validation. We validate the A -only certificate against independently assembled sweep operators and observed IBMI trajectories, and then investigate the effects of overlap and ordering. The experiments focus on certificate accuracy, agreement with the observed convergence dynamics, and the structural sensitivity of the IBMI convergence factor to block geometry and ordering.

9.1. Covariance protocol and direct certificate check. To remain comparable with the original IBMI study [16], we use its EXP, RBF, inverse-quadratic, Matérn-3/2, and Matérn-5/2 covariance families. We freeze $\sigma = 0.5$ for RBF and $\tau = 3$ for the Matérn kernels as representative experimental choices. One-dimensional points are equally spaced on $[0, n^{0.9}]$; two-dimensional points form a regular grid on the unit square. We use $K = 4$ contiguous blocks with nominal 5% overlap and dimensions $n = 64, 128, 256$ in one dimension and $n = 64, 144, 256$ in two dimensions.

For all 30 instances, we compute $\gamma_A = \rho(M_A)^2$ from local SPD solves. On the 20 cases with $n \leq 144$, independent assembly of Q gives a maximum absolute $\log_{10}$ discrepancy between γ_A and $\rho(Q)^2$ of 1.53×10^{-12} when $\kappa_2(A) < 10^{12}$ and 2.29×10^{-8} over all cases.

9.2. Actual trajectories and public SPD matrices. We distinguish the exact certificate γ_A , the independently propagated congruence $E_{s+1} = QE_sQ^\top$, and actual IBMI block reconstruction. Rate estimates use only errors safely above a conditioning-aware floating-point floor.

As an external check, we use the eight public stiffness matrices HB/bcsstk01–HB/bcsstk08 from the SuiteSparse Matrix Collection [3]. Each matrix passes an SPD Cholesky check. We apply diagonal congruence scaling before the experiment and use $K = 4$ contiguous blocks with nominal overlap 0%, 5%, 10%, 20%. For each

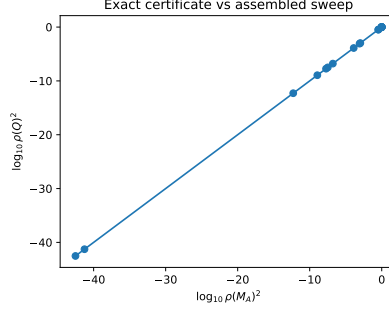


FIG. 9.1. *Exact A-only certificate versus the spectral factor of an independently assembled sweep operator. The diagonal is equality.*

TABLE 9.1

Native-order exact certificate γ_A on the scaled public SPD matrices. The final column is the factor ratio between 0% and 20% overlap.

matrix	0%	5%	10%	20%	ratio
bsstk01	.993623	.993467	.993467	.993467	1.000
bsstk02	.986422	.978287	.971695	.940730	1.049
bsstk03	.983275	.891011	.678239	.048749	20.170
bsstk04	.985801	.980258	.974219	.937667	1.051
bsstk05	.987576	.977014	.959700	.914594	1.080
bsstk06	.996790	.994596	.991774	.986704	1.010
bsstk07	.996790	.994596	.991774	.986704	1.010
bsstk08	.996312	.995799	.989490	.970852	1.026

matrix/overlap pair, we run native, one frozen random, and certificate-minimizing orders, giving 96 real-matrix trajectories.

All 96 trajectories have an observable pre-floor regime. The median relative discrepancy between γ_A and the propagated pre-floor factor is 5.87×10^{-7} (90th percentile 6.56×10^{-5}); using the final observable local operator ratio gives 5.78×10^{-7} and 6.44×10^{-5} , respectively. Actual IBMI and propagated congruence trajectories have median within-trajectory discrepancy 1.40×10^{-12} above the numerical floor.

Table 9.1 reports the certificate for the native ordering. The observed effect of overlap varies substantially across matrices. The most pronounced case is **bsstk03**, for which the certificate decreases from 0.9833 at 0% overlap to 0.04875 at 20%. The corresponding trajectory behavior shows the same qualitative trend: the 0% case does not reach the prescribed tolerance within the 300-sweep limit, whereas the 5%, 10%, and 20% cases require 150, 45, and 7 sweeps, respectively. These observations illustrate the potential impact of overlap for a particular matrix and ordering, but do not imply monotonicity for general multi-block covers. Indeed, Theorem 6.2 shows that the two-block monotonicity property can already fail for a three-block ordered cover.

A finite-precision guardrail appears for the two-dimensional Matérn-5/2 case at $n = 256$: with 20% overlap, $\gamma_A \approx 0.0911$ and operator ratios are 0.088–0.096, while reconstruction stalls near 10^{-7} at $\kappa_2(A) \approx 2.1 \times 10^{11}$.

9.3. Ordering sensitivity and certificate-guided ordering. For the public matrices with contiguous blocks, the effect of ordering is negligible: across the 32 matrix–overlap pairs, the median ratio between the native-order and best certificate values is 1.000000, and the maximum is 1.000517. Thus, the existence of an ex-

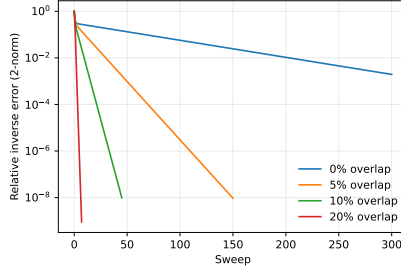


FIG. 9.2. Actual IBMI relative inverse-error trajectories for `bcsstk03` under native ordering. Increasing overlap strongly changes this particular instance, consistent with the exact certificates in Table 9.1.

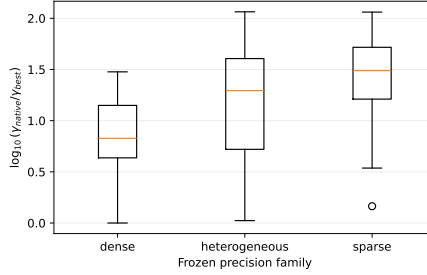


FIG. 9.3. Ordering sensitivity in the frozen general fixed-block stress test. Each box contains all 20 seeds in the indicated precision-matrix regime; the vertical quantity is the native-to-best exact certificate ratio on a $\log_{10}$ scale.

act ordering objective does not imply substantial ordering sensitivity for every block family.

To examine ordering sensitivity under controlled conditions, we consider a separate stress-test ensemble of SPD precision matrices with unit diagonal and strict diagonal dominance. The ensemble comprises three coupling regimes (dense, heterogeneous, and sparse) with 20 fixed random seeds per regime. For each instance, we use a fixed-block family with $n = 48$ and $K = 6$ and evaluate all $6! = 720$ permutations exactly using cached A -only local maps. These instances are constructed by prescribing the precision matrix H and then forming $A = H^{-1}$; accordingly, they serve as controlled tests of the ordering mechanism rather than as a statistical model for the prevalence of ordering sensitivity in application matrices.

The median native-to-best certificate ratios are 6.76, 19.66, and 31.09 in the dense, heterogeneous, and sparse regimes, respectively. Across all 60 instances, 90% have a native-to-best ratio of at least two, 76.7% of at least five, and 61.7% of at least ten; the largest observed ratio is 115.76. Figure 9.3 reports the results for the complete ensemble, including instances with weak ordering sensitivity.

The singleton-complement specialization provides a sharper structural check. For $n = 8$ we enumerate all $8! = 40320$ orders. In the dense and heterogeneous precision families, the median native-to-optimal ratios are 1.58×10^4 and 2.71×10^8 , respectively. In the sparse family, zero precision couplings produce zero cycle products for some orders; these are finite-termination instances of the graph mechanism in Section 8, so ratios with zero optimum are not used in aggregate improvement statistics. A deterministic 2-opt heuristic reaches the exhaustive optimum in 60% of the 40

643 dense/heterogeneous cases and has median optimum ratio numerically equal to one,
 644 but its worst ratio is 4.89×10^3 . We therefore make no approximation guarantee for
 645 2-opt.

646 **9.4. Cost, scope, and reproducibility.** For the covariance campaign with
 647 $K = 4$, cached and uncached evaluations of all 24 permutations agree to the recorded
 648 precision. Across six representative cases, caching yields a median scoring speedup
 649 of $3.48\times$, or $3.12\times$ when cache construction is included. These hardware-dependent
 650 timings demonstrate the computational benefit of factor-once reuse for a fixed block
 651 collection. By contrast, in the tested regime, screening alternative block covers did
 652 not amortize the additional cost of constructing their certificates.

653 Overall, the experiments confirm the accuracy of the certificate for the persistent
 654 IBMI dynamics and show that the effects of overlap and ordering depend strongly on
 655 the matrix and block geometry. Complete experimental protocols, raw results, and
 656 reproduction scripts are provided in the accompanying repository.

657 *Reproducibility.* The code and scripts used to reproduce the numerical experi-
 658 ments, including the covariance validation, public-SPD experiments, and ordering-
 659 stress tests, are publicly available in the [accompanying GitHub repository](#).

660 **10. Discussion.** Theorem 5.2 makes the persistent sweep spectrum computable
 661 from A and the ordered cover without forming the full inverse. It thereby turns the
 662 return representation into a pre-iteration convergence certificate and makes overlap
 663 and ordering accessible as design variables. The scope of these conclusions is precise.
 664 Nested overlap is monotone in the two-block setting, whereas this property can fail
 665 already for a three-block ordered cover. For a fixed block collection, cached local graph
 666 maps enable repeated evaluation of different orderings without recomputing the local
 667 factorizations. The certificate characterizes the asymptotic spectral factor exactly,
 668 while finite-sweep behavior may still exhibit the nonnormal transients described in
 669 Proposition 3.4.

670 The numerical results reinforce these distinctions. On the public SPD stiffness
 671 matrices, the certificate accurately captures the persistent dynamics, while the tested
 672 contiguous block families exhibit only weak ordering sensitivity. In contrast, the
 673 controlled ordering-stress experiments demonstrate that substantial ordering effects
 674 can arise under heterogeneous transfer geometry. Thus, ordering sensitivity depends
 675 on the interaction between the matrix and block geometry rather than on the existence
 676 of an ordering objective alone. Developing approximation guarantees for the NP-hard
 677 rate-optimal ordering problem remains an open direction.

678 **11. Conclusion.** Overlapping IBMI admits a structured congruence error dy-
 679 namics whose persistent spectrum is governed by a return operator computable exactly
 680 from local solves with the original SPD matrix A . Consequently, $\rho(M_A)^2$ provides an
 681 exact pre-iteration asymptotic spectral certificate for the full inverse-error operator.
 682 The same framework yields a monotonicity law for nested overlap in the two-block set-
 683 ting, together with its three-block limitation; an exact precision-cycle objective whose
 684 rate-optimal ordering problem is NP-hard; and a factor-once, score-many procedure
 685 for evaluating orderings of a fixed block collection.

686 *Declaration on the use of artificial intelligence.* Generative AI tools were used to
 687 assist with code development and verification, as well as language editing, during the
 688 preparation of this manuscript. All mathematical statements, proofs, computational
 689 results, references, and conclusions were independently reviewed and verified by the
 690 author. The author assumes responsibility for all content.

REFERENCES

- [1] M. BENZI, A. FROMMER, R. NABBEN, AND D. B. SZYLD, *Algebraic theory of multiplicative schwarz methods*, Numerische Mathematik, 89 (2001), pp. 605–639, <https://doi.org/10.1007/s002110100275>.
- [2] Å. BJÖRCK AND G. H. GOLUB, *Numerical methods for computing angles between linear subspaces*, Mathematics of Computation, 27 (1973), pp. 579–594, <https://doi.org/10.1090/S0025-5718-1973-0348991-3>.
- [3] T. A. DAVIS AND Y. HU, *The university of florida sparse matrix collection*, ACM Transactions on Mathematical Software, 38 (2011), pp. 1:1–1:25, <https://doi.org/10.1145/2049662.2049663>.
- [4] A. P. DEMPSTER, *Covariance selection*, Biometrics, 28 (1972), pp. 157–175, <https://doi.org/10.2307/2528966>.
- [5] Z. FRANGELLA, J. A. TROPP, AND M. UDELL, *Randomized nystrom preconditioning*, SIAM Journal on Matrix Analysis and Applications, 44 (2023), pp. 718–752, <https://doi.org/10.1137/21M1466244>.
- [6] M. J. GANDER, *Optimized schwarz methods*, SIAM Journal on Numerical Analysis, 44 (2006), pp. 699–731, <https://doi.org/10.1137/S0036142903425409>.
- [7] M. J. GANDER, S. LOISEL, AND D. B. SZYLD, *An optimal block iterative method and preconditioner for banded matrices with applications to PDEs on irregular domains*, SIAM Journal on Matrix Analysis and Applications, 33 (2012), pp. 653–680, <https://doi.org/10.1137/100796194>.
- [8] M. R. GAREY, D. S. JOHNSON, AND R. E. TARJAN, *The planar hamiltonian circuit problem is NP-complete*, SIAM Journal on Computing, 5 (1976), pp. 704–714, <https://doi.org/10.1137/0205049>.
- [9] R. M. GOWER AND P. RICHTÁRIK, *Randomized quasi-newton updates are linearly convergent matrix inversion algorithms*, SIAM Journal on Matrix Analysis and Applications, 38 (2017), pp. 1380–1409, <https://doi.org/10.1137/16M1062053>.
- [10] M. GRIEBEL AND P. OSWALD, *Greedy and randomized versions of the multiplicative schwarz method*, Linear Algebra and its Applications, 437 (2012), pp. 1596–1610, <https://doi.org/10.1016/j.laa.2012.04.052>.
- [11] X. HU, J. XU, AND L. T. ZIKATANOV, *Randomized and fault-tolerant method of subspace corrections*, Research in the Mathematical Sciences, 6 (2019), p. 29, <https://doi.org/10.1007/s40687-019-0187-z>.
- [12] S. HUAN, J. GUINNESS, M. KATZFUSS, H. OWHADI, AND F. SCHÄFER, *Sparse inverse cholesky factorization of dense kernel matrices by greedy conditional selection*, SIAM/ASA Journal on Uncertainty Quantification, 13 (2025), pp. 1649–1679, <https://doi.org/10.1137/23M1606253>.
- [13] L. LIN, C. YANG, J. LU, L. YING, AND W. E, *A fast parallel algorithm for selected inversion of structured sparse matrices with application to 2d electronic structure calculations*, SIAM Journal on Scientific Computing, 33 (2011), pp. 1329–1351, <https://doi.org/10.1137/09077432X>.
- [14] P. OSWALD AND W. ZHOU, *Random reordering in SOR-type methods*, Numerische Mathematik, 135 (2017), pp. 1207–1220, <https://doi.org/10.1007/s00211-016-0829-7>.
- [15] P. OSWALD AND W. ZHOU, *Correction: Random reordering in SOR-type methods*, Numerische Mathematik, 154 (2023), pp. 521–525, <https://doi.org/10.1007/s00211-023-01365-9>.
- [16] A. PATERSON, J. PESTANA, AND V. DOLEAN MAINI, *An iterative block matrix inversion (IBMI) algorithm for symmetric positive definite matrices with applications to covariance matrices*, SIAM Journal on Matrix Analysis and Applications, 47 (2026), pp. 1605–1627, <https://doi.org/10.1137/25M1732076>.
- [17] F. SCHÄFER, M. KATZFUSS, AND H. OWHADI, *Sparse cholesky factorization by kullback–leibler minimization*, SIAM Journal on Scientific Computing, 43 (2021), pp. A2019–A2046, <https://doi.org/10.1137/20M1336254>.
- [18] F. SCHÄFER, T. J. SULLIVAN, AND H. OWHADI, *Compression, inversion, and approximate PCA of dense kernel matrices at near-linear computational complexity*, Multiscale Modeling & Simulation, 19 (2021), pp. 688–730, <https://doi.org/10.1137/19M129526X>.
- [19] J. XU, *Iterative methods by space decomposition and subspace correction*, SIAM Review, 34 (1992), pp. 581–613, <https://doi.org/10.1137/1034116>.
- [20] J. XU AND L. ZIKATANOV, *The method of alternating projections and the method of subspace corrections in Hilbert space*, Journal of the American Mathematical Society, 15 (2002), pp. 573–597, <https://doi.org/10.1090/S0894-0347-02-00398-3>.